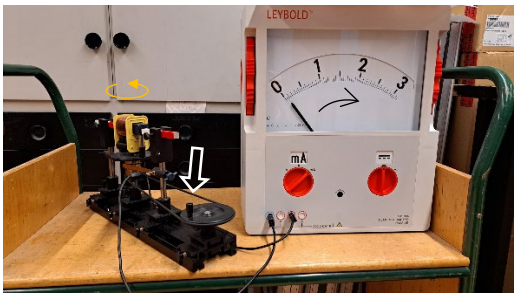
Electromagnetism for a sustainable future

The exercises of this week illustrate electromagnetism in use for the green transition. These include electricity generated by a wind turbine and the motor driving an electric vehicle.

Generation of electricity: from mechanical motion to electrical current

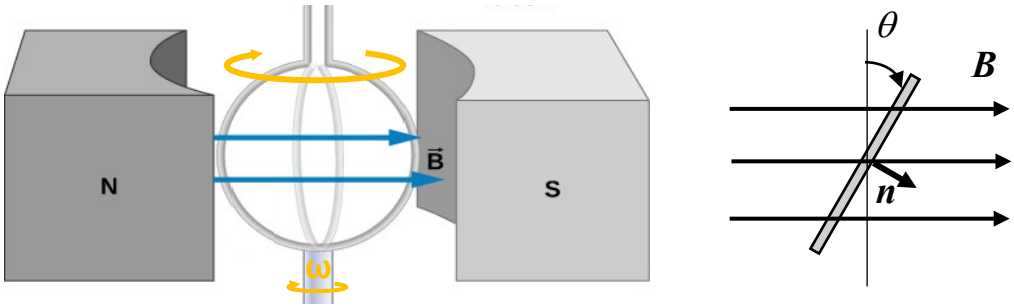
At the lecture you may have seen a demonstration of the generator shown in the photo to the right: When the handle (white arrow) is turned, the solenoid with copper turns is turned in the magnetic field between the two permanent magnets, and a current is generated.

In the following two exercises you will work more on motional emf, which is used for the generation of current in wind turbines and bicycle lamps, respectively.



Example of generator for electricity (motional emf). When the solenoid is rotated in the magnetic field between the two permanent magnets a current is generated.

Exercise 1

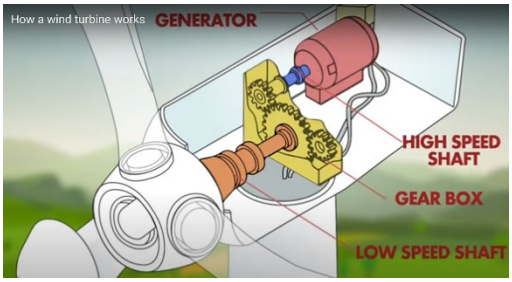


The figure above shows a solenoid that is being turned in a homogenous B-field similar to the demonstration experiment. The solenoid has current leads at the top and can rotate around the shaft at the bottom. This shaft could be inside a wind turbine. The solenoid is rotated with constant angular speed ω via the shaft. The radius of the solenoid is r. The current leads are connected via a resistance (not shown) of magnitude R_0. The figure to the right shows the solenoid as seen from above with rotation angle θ and normal vector $\boldsymbol{n}$. At time t=0 the normal vector is parallel to the B-field.

a) Determine the flux of the magnetic field through the solenoid as function of the rotation angle θ, with the sign given by the normal vector.

b) Determine the induced emf and the current in the solenoid as function of time. Assume that magnetic fields generated by current in the solenoid (self-induction) can be neglected in

comparison with the external *B*-felt. Also assume that the resistance of the solenoid itself can be neglected.



For larger wind turbines the blades rotate relatively slower. Therefore a gear box is often used between the shaft for the blades (low speed shaft) and the generator (high speed shaft), so ω becomes of a significant magnitude (e.g. 1500 rpm), image from [G1].

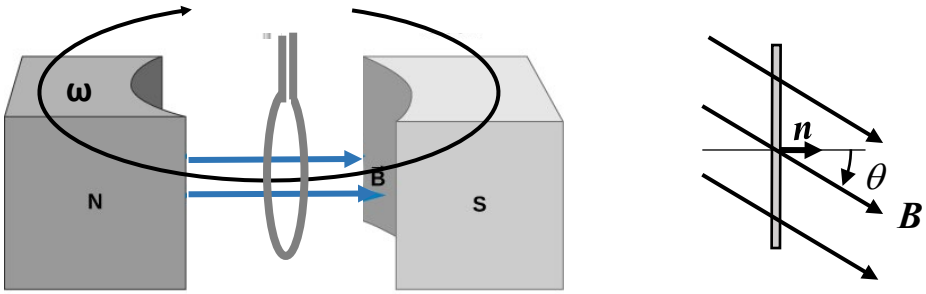
c) Discuss what can be done to increase the induced emf in the solenoid, which parameters are of importance for the emf and how/how much can these realistically be improved.

d) Determine the torque on the solenoid caused by the magnetic field and the induced current. Be careful with the sign – in which direction will the torque act on the solenoid?

e) If the angular speed should be kept constant, it implies that the torque determined in c) should be counteracted by the torque from the shaft, i.e., this must be varying. A wind turbine will naturally deliver a constant torque. Discuss how this problem can be solved or diminished.

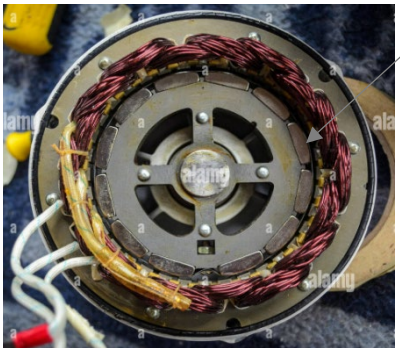
In the examples above, the solenoid is turned while the magnets are fixed. In a generator it is often the solenoid that is fixed and the magnets that are rotating.

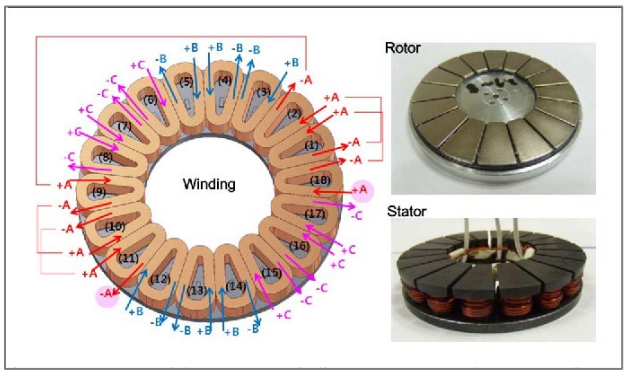
f) Show that the resulting emf and current in the solenoid and the torque on the solenoid become the same (show it following the method in question a)-e) but with reference to the situation skethed in the figure below where the magnets and not the solenoid are turning). Discuss what the practical implications are if the magnets are rotating in stead of the solenoid.



Generators in wind turbines:

Here are shown some examples of how solenoids and magnets are configured in a wind turbine generator:

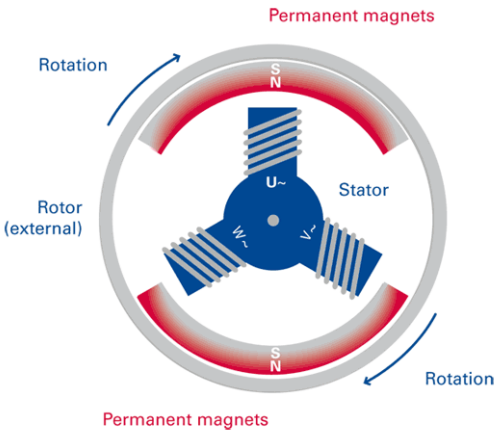




From [MG1]

The magnets are typically alloys of NdFeB. They are presently the strongest permanent magnets available, and they can be fabricated in a flat geometry without demagnetizing, which provides good design opportunities. The NdFeB magnets were developed in the 1990'ies. Today, research is focused on developing even stronger permanent magnets, preferably without the use of rare earth elements like Nd. The stronger magnets, the more electricity can be produced by the wind turbine. Approximately 1 ton magnetic material is used for a megawatt (MW) wind turbine.





Small generators: bicycle light without batteries

25 years ago a young student at DTU (Institute for Product Development), Troels Petersen, invented a new type of bicycle light [C1, C2]. The light is driven by a magnet on the wheel. The magnet passes a solenoid mounted on the bicycle frame. The solenoid produces electricity for a diode light. Troels Petersen filed a patent on the idea [P1, P2] and he later sold the rights to the company Reelight. Some of you may even have his lights on your bicycle (see the left photo).

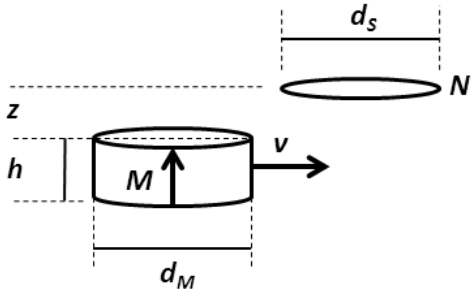




Magnet bicycle light.
At the rigth photos the parts have been diassembled, so the copper solenoid, diodes and the magnets can be seen. The magnets are flat NdFeB magnets.

Here we will here calculate an example for designing a magnet bicycle light:

Exercise 2
A copper wire solenoid is mounted on a bicycle frame. The solenoid has diameter $d_S = 1$ cm and $N = 1000$ turns. A magnet with the shape of a cylinder is mounted on the wheel. The magnet has diameter $d_M = 1$ cm, height $h = 0.5$ cm and the magnetization $M = 1000$ kA/m. It passes the solenoid with the speed $v = 18$ km/h as shown in the figure below. During passing the distance between magnet and solenoid is $z = 2.5$ mm. Assume that the magnetic field in the solenoid changes linearly from $B = 0$ until its maximum during the time it takes the magnet to move the distance d_M.



Simplified sketch of an induction solenoid and magnet for a bicycle light.

Assume that the magnet can be modelled as a single circular current loop in its symmetry plane. The B-field on the symmetry axis in the distance z from the magnet surface can be written as:

$$\mathbf{B}(z) = \frac{\mu_0 M h d_M^{\,2}}{8\left[\left(\frac{d_M}{2}\right)^2 + \left(\frac{h}{2} + z\right)^2\right]^{\frac{3}{2}}}.$$

a) Determine the order of magnitude for the induced voltage in the solenoid. Assume that the given magnetic field can be used for the entire area of the solenoid.

Discuss if the given values for the parameters (for example v og z) make sense. If not, what should they be, and how much does this change the induced voltage?

Two diode lights are mounted inside the bicycle light. Each diode lights at a minimum voltage and current of 2.5 V and 20 mA, respectively. The diodes and the solenoid are connected in series. You can disregard the self-inductance of the solenoid.

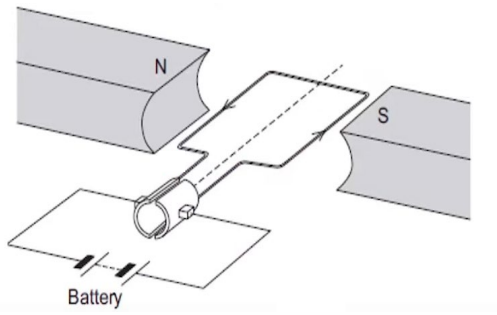
b) What should the diameter of the copper wire in the solenoid at least be for the diodes to light?

Electric motors

The working principle of electric motors is the opposite of generators: they convert electric current into mechanical motion. There is a large variety of applications, e.g. power drills and electric vehicles, just to mention a few.

Exercise 3

In chapter 11.5 (OpenStax) was presented the principle of a DC -motor (a direct current motor). A current runs through the loop in a magnetic field as shown in the figure below.



a) The magnetic force will turn the loop. Write the equation for this force. Determine the direction of the force on each of the wire sections of the loop. Which direction will the loop turn?

b) At the connection for the loop a commutator is placed, so the current changes direction every time the plane of loop is perpendicular to the magnetic field. Why is this commutator important?

c) How could you improve this motor? For instance, consider if the torque on the solenoid is constant during the turning. If not, where is the torque largest and smallest and how does this variation affect the rotation? What could be done to reduce the variation in the torque? What could be done to increase the torque on the solenoid?

d) The most important element in an electric motor is that an applied current results in a force that mechanically can drive a shaft. The electric motor in this exercise/OpenStax chapter 11.5 is the most shown example in textbooks. However, an electric motor can be designed in many other ways. Suggest at least two other ways that would result in a force created by an applied current and/or magnets.

This video explains how the DC-motor in chapter 11.5 works: DC EM - Text book

This video explains how a real DC-motor works: DC EM - Real stuff

e) At the lecture you may have seen an ”ideal Ørsted motor” and a more practical motor with more solenoids. Explain the physics principles behind the latter. Is this a direct current or an alternating current (DC or AC) motor? Where could this type of motor be applied?

f) In electric vehicles of the brand Tesla an induction motor is used [IM1]. The induction motor was invented by Nikola Tesla in 1887 [IM2]. This is an AC-motor. I works according to this principle: An AC-current in the stationary part of the motor (the stator) creates a time-varying magnetic field that then induces a current in the rotor, which results in a torque that makes the rotor rotate. It doesn’t need magnets to function. This video shows the principle behind an induction motor: How does an Induction Motor work. Explain the principle.